

Design and Simulation of an Automatic Street Light Control System Using Combinational Logic

A Group Project in Digital Logic Design — Combinational Circuit Application for Smart Urban Infrastructure

Introduction: Automatic Street Lighting in Smart Cities

What is an Automatic Street Light?

A lighting system that turns **ON/OFF based on environmental conditions** — such as daylight levels and motion — without any manual intervention.

Smart City Integration

Automatic street lights reduce energy consumption by **40–60%** compared to traditional always-on systems, making them a cornerstone of sustainable urban infrastructure.

Project Goal

Design and simulate a **combinational logic circuit** that automates street lighting using basic digital logic gates (AND, OR, NOT).



Energy Efficiency

Lights activate only when needed — never during daylight hours



Cost Savings

Lower electricity bills for municipalities across the city



Sustainability

Reduced carbon footprint in urban infrastructure



Safety

Improved visibility automatically when motion is detected

Problem Statement and Objectives

The Problem

Traditional street lights operate on **fixed schedules or manual control**, wasting energy during low-traffic periods and daylight hours.

Project Objectives

1 Design a combinational logic circuit

Using basic gates: AND, OR, NOT

2 Implement sensor inputs

Daylight detection (LDR) and Motion detection

3 Create a complete truth table

Defining all 8 operational scenarios

4 Simulate and validate

Verify energy reduction while maintaining safety

Daytime Waste

Unnecessary lighting active during daytime hours

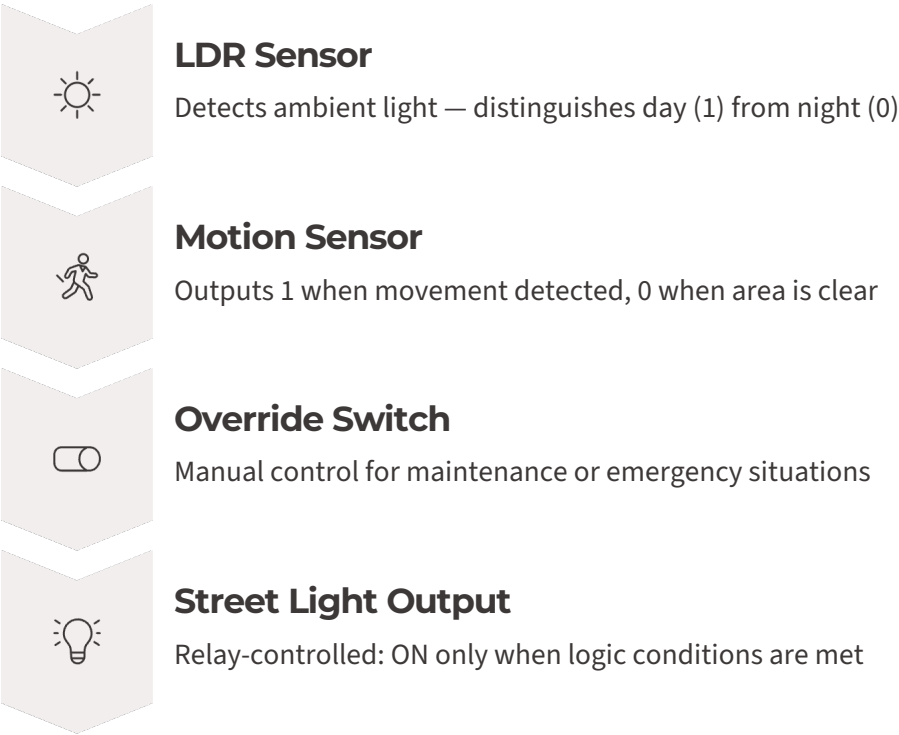
Always-On

Continuous operation even when no one is present

Budget Burden

Significant cost pressure on city lighting budgets

Working Principle: How the System Operates



 Daytime (D=1) Light remains OFF regardless of motion — energy saved	 Night + No Motion Light stays OFF — no one present, power conserved
 Night + Motion Light turns ON automatically — safety guaranteed	 Override (O=1) Light turns ON regardless — maintenance mode active

Logic & Truth Table: Combinational Logic Breakdown

Logic Expression

$$L = O + (\neg D \cdot M)$$

The light turns **ON** if:

- **Override (O)** is active, *OR*
- It is **nighttime ($\neg D$)** AND **motion (M)** is detected

Inputs: O = Override | D = Daylight | M = Motion

Output: L = Street Light (1 = ON, 0 = OFF)



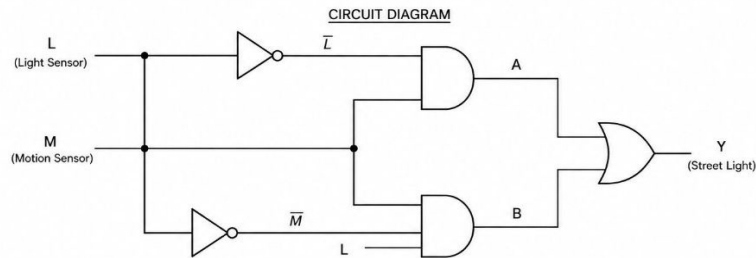
Only 2 out of 8 combinations activate the light without Override — proving the system's energy efficiency.

O	D	M	L	Scenario
0	0	0	0	Nighttime, no motion → OFF
0	0	1	1	Night + Motion → ON
0	1	0	0	Daytime, no motion → OFF
0	1	1	0	Daytime + Motion → OFF
1	0	0	1	Override ON → Light ON
1	0	1	1	Override ON → Light ON
1	1	0	1	Override ON → Light ON
1	1	1	1	Override ON → Light ON

Circuit Design: Components and Architecture

Automatic Street Light Control System

Using Combinational Logic



HOW IT WORKS

- The street light turns ON when it is DARK ($L = 0$), regardless of motion.
- The street light turns ON when motion is detected ($M = 1$), even if it is BRIGHT ($L = 1$).
- The street light turns OFF only when it is BRIGHT ($L = 1$) and NO MOTION is detected ($M = 0$).

LOGIC EXPRESSION

$$Y = (\bar{L} \cdot M) + (L \cdot \bar{M}) = A + B$$

LEGEND

- L → Light Sensor (LDR)
- M → Motion Sensor
- Y → Street Light (Output)
- 1 → HIGH (ON)
- 0 → LOW (OFF)

Primary Components Signal Flow

1. LDR → **NOT gate** (inverts daylight signal)
2. $\neg D + M \rightarrow$ **AND gate** (night-and-motion check)
3. AND output + O → **OR gate** (final decision)
4. OR output → **Relay** → Street Light

LDR Sensor

Converts ambient light to digital signal — 0 = night, 1 = day

Motion Sensor

Outputs 1 on movement, 0 when area is still

Override Switch

Manual input — activates maintenance mode

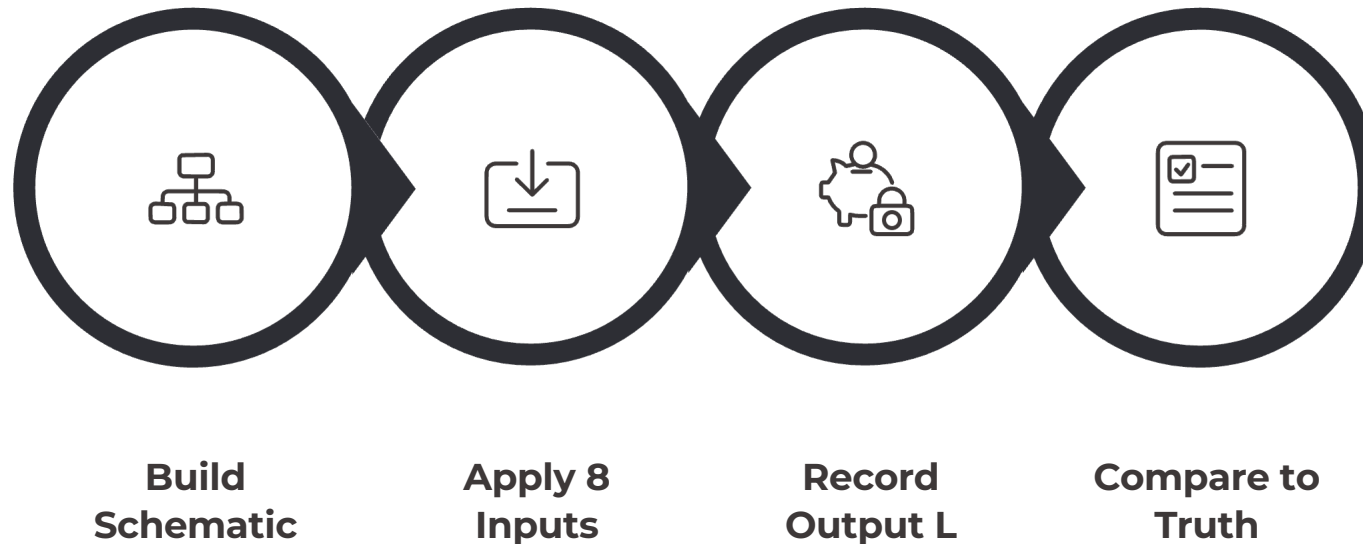
Logic Gates

NOT, AND, OR gates implementing $L = O + (\neg D \cdot M)$

Relay

Switches the street light ON/OFF based on logic output

Simulation: Process, Software, and Virtual Testing



Simulation Objective

Verify that the combinational logic circuit behaves correctly for **all 8 input combinations** before physical implementation.

Software Tools

Virtual Environment Benefits

- No risk of hardware damage during testing
- Easy modification of circuit parameters
- Rapid iteration and debugging
- Cost-effective validation before physical build

Proteus / Multisim / LTspice

Full circuit schematic simulation and virtual testing

Logic Simulation

Truth table verification and timing analysis

Waveform Analysis

Observe input/output signal behavior over time

Results & Real-World Integration: From Theory to Smart Cities

✓ **All 8 Cases Matched**

Simulated outputs matched the truth table perfectly across every input combination

✓ **Logic Validated**

$L = O + (\neg D \cdot M)$ confirmed successfully in virtual environment

✓ **Timing Verified**

Gate delays within acceptable limits for real-world street light control

Real-World Benefits

- **Energy Savings**
40–60% reduction in electricity consumption vs. always-on systems
- **Cost Reduction**
Lower municipal budgets for street lighting operations city-wide
- **Sustainability**
Reduced carbon emissions and environmental impact
- **Smart City Scalability**
Deployable across thousands of lights with IoT connectivity for centralized monitoring

Future Enhancements

- 01
Weather Sensor Integration
Adapt lighting to fog, rain, or storm conditions automatically
- 02
Traffic Data Fusion
Correlate street light activation with real-time traffic flow
- 03
Cloud-Based Scheduling
Centralized dashboard for remote monitoring and control at scale

❏ **Proof of Concept:** Combinational logic successfully automates street light control — eliminating manual operation and reducing unnecessary energy consumption in smart cities.